Chapter 15, Problem 38P

Problem

Find $f(t)$ given that:

(a) $F(s) = \dfrac{s^2 + 4s}{s^2 + 10s + 26}$

(b) $F(s) = \dfrac{5s^2 + 7s + 29}{s(s^2 + 4s + 29)}$

Step-by-step solution

Step 1 of 5

(a)

Consider the following transfer function:

$$\begin{aligned} F(s) &= \frac{s^2+4s}{s^2+10s+26} \\ &= \frac{s^2+4s+6s-6s+26-26}{s^2+10s+26} \\ &= \frac{s^2+10s+26-6s-26}{s^2+10s+26} \\ &= \frac{s^2+10s+26}{s^2+10s+26} - \frac{6s+26}{s^2+10s+26} \end{aligned}$$

$$F(s) = 1 - \frac{6s+26}{s^2+10s+26} \quad \ldots\ldots (1)$$

Step 2 of 5

Furhter simplification of equation (1) is as follows.

$$\begin{aligned} F(s) &= 1-\frac{6s+26}{s^2+10s+26} \\ &= 1-\frac{6s+26}{(s+5)^2+1} \\ &= 1-\left(\frac{6s+30-4}{(s+5)^2+1}\right) \\ &= 1-\frac{6s+30}{(s+5)^2+1}+\frac{4}{(s+5)^2+1} \end{aligned}$$

$$F(s)=1-6\left(\frac{s+5}{(s+5)^2+1}\right)+4\left(\frac{1}{(s+5)^2+1}\right)$$

Apply inverse laplace transformation on both sides,

$$\mathcal{L}^{-1}[F(s)]=\mathcal{L}^{-1}\left[1-6\left(\frac{s+5}{(s+5)^2+1}\right)+4\left(\frac{1}{(s+5)^2+1}\right)\right]$$

$$\mathcal{L}^{-1}[F(s)]=\mathcal{L}^{-1}[1]-6\mathcal{L}^{-1}\left[\frac{s+5}{(s+5)^2+1}\right]+4\mathcal{L}^{-1}\left[\frac{1}{(s+5)^2+1}\right]$$

$$f(t)=\delta(t)-6e^{-5t}\cos t\,u(t)+4e^{-5t}\sin t\,u(t)$$

$$f(t)=\delta(t)-6e^{-5t}\left[\cos t-\frac{2}{3}\sin t\right]u(t)$$

Therefore, the Laplace transform of the function $F(s)$ is,

$$\boxed{f(t)=\delta(t)-6e^{-5t}\left[\cos t-\frac{2}{3}\sin t\right]u(t)}.$$

Step 3 of 5

(b)

Consider the following transfer function:

$$F(s)=\frac{5s^2+7s+29}{s(s^2+4s+29)}$$

Use partial fractions to simplify the function $F(s)$.

$$\frac{5s^2+7s+29}{s(s^2+4s+29)}=\frac{A}{s}+\frac{Bs+C}{s^2+4s+29} \quad \text{...... (2)}$$

$$5s^2+7s+29=A(s^2+4s+29)+(Bs+C)s$$

$$5s^2+7s+29=A(s^2+4s+29)+Bs^2+Cs \quad \text{...... (3)}$$

Compare the coefficient of constant terms in the equation (3).

$$\begin{aligned} 29A &= 29 \\ A &= 1 \end{aligned}$$

Compare the coefficient of s^2 terms in the equation (3).

$$\begin{aligned} A+B &= 5 \\ 1+B &= 5 \\ B &= 4 \end{aligned}$$

Compare the coefficient of s terms in the equation (3).

$$\begin{aligned} 4A+C &= 7 \\ 4(1)+C &= 7 \\ C &= 3 \end{aligned}$$

Substitute 1 for A, 4 for B and 3 for C in equation (2).

$$\frac{5s^2+7s+29}{s\left(s^2+4s+29\right)}=\frac{1}{s}+\frac{4s+3}{s^2+4s+29}$$

$$\begin{aligned}F(s)&=\frac{1}{s}+\frac{4s+3}{s^2+4s+4+25}\\&=\frac{1}{s}+\frac{4s+3}{(s+2)^2+5^2}\\&=\frac{1}{s}+\frac{4s+8-8+3}{(s+2)^2+5^2}\end{aligned}$$

$$F(s)=\frac{1}{s}+4\frac{s+2}{(s+2)^2+5^2}-\frac{5}{(s+2)^2+5^2} \quad \text{...... (4)}$$

Apply inverse laplace transformation to equation (4),

$$\mathcal{L}^{-1}\left[F(s)\right]=\mathcal{L}^{-1}\left[\frac{1}{s}+4\left(\frac{s+2}{(s+2)^2+5^2}\right)-\frac{5}{(s+2)^2+5^2}\right]$$

$$\mathcal{L}^{-1}\left[F(s)\right]=\mathcal{L}^{-1}\left[\frac{1}{s}\right]+4\mathcal{L}^{-1}\left[\frac{s+2}{(s+2)^2+5^2}\right]-\mathcal{L}^{-1}\left[\frac{5}{(s+2)^2+5^2}\right]$$

$$f(t)=u(t)+4e^{-2t}\cos 5t\,u(t)-e^{-2t}\sin 5t\;u(t)$$

$$f(t)=\left\{1+e^{-2t}\left[4\cos 5t-\sin 5t\right]\right\}u(t)$$

Therefore, the Laplace transform of the function $F(s)$ is,

$$\boxed{f(t)=\left\{1+e^{-2t}\left[4\cos 5t-\sin 5t\right]\right\}u(t)}.$$